Parallel and distributed Computing

Lecture 16

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Load Balancing

Introduction

- Processing speed of a system is always highly intended.
- Distributed computing system provides high performance environment that are able to provide huge processing power.
- In distributed computing thousand of processors can be connected either by wide area network or across a large number of systems which consists of cheap and easily available autonomous systems like workstations or PCs.
- The distribution of loads to the processing elements is simply called the **load balancing**.
- The **goal** of the **load balancing algorithms** is to maintain the load to each processing element such that all the processing elements become neither overloaded nor idle that means each processing element ideally has equal load at any moment of time during execution to obtain the maximum performance (minimum execution time) of the system.

Load Balancing

- **Load balancing** is the way of distributing load units (jobs or tasks) across a set of processors which are connected to a network which may be distributed across the globe.
- The excess load or remaining unexecuted load from a processor is migrated to other processors which have load below the threshold load.
- **Threshold load** is such an amount of load to a processor that any load may come further to that processor.
- By load balancing strategy it is possible to make every processor equally busy and to finish the works approximately at the same time.

Load Balancing Operation

A Load Balancing Operation is defined by three rules:

Location Rule

- This rule determines the partners of balancing operation i.e. The processors to involve in the balancing operation.

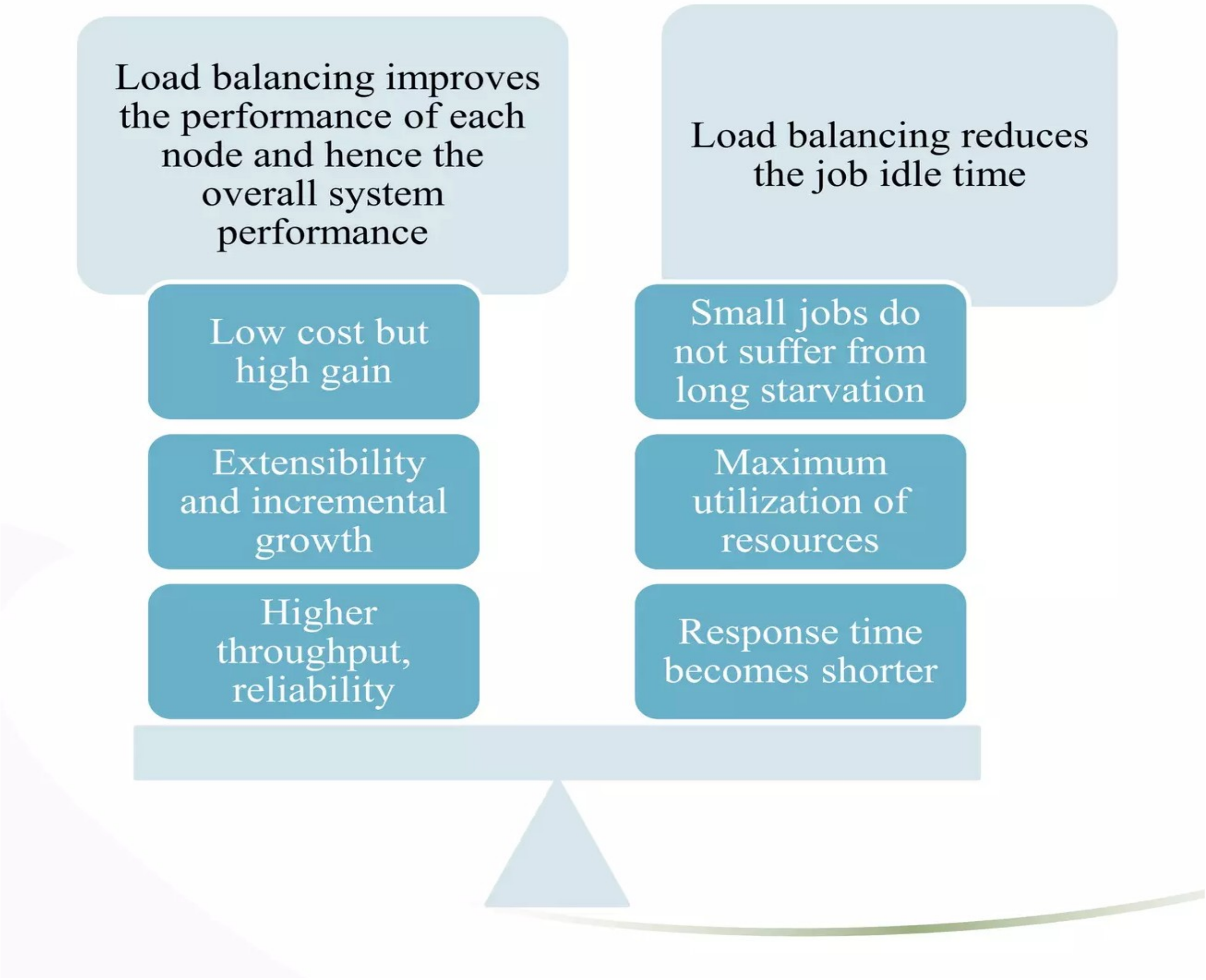
Distribution Rule

- This rule determines how to redistribute workload among the processors in the balancing domain.

Selection Rule

- The selection rule works either in preemptive or in nonpreemptive
- fashion.

Benefits of load balancing



Static load balancing

- In static algorithm the processes are assigned to the processors **at the compile time** according to the performance of the nodes.
- Once the processes are assigned, **no change or reassignment** is possible at the run time.
- Number of jobs in each node is fixed in static load balancing algorithm. Static algorithms do not collect any information about the nodes.
- The assignment of jobs is done to the processing nodes on the basis of the following factors: **incoming time, extent of resource needed, mean execution time and inter-process communications.**
- Since these factors should be measured before the assignment, this is why static load balance is also called **probabilistic algorithm**.

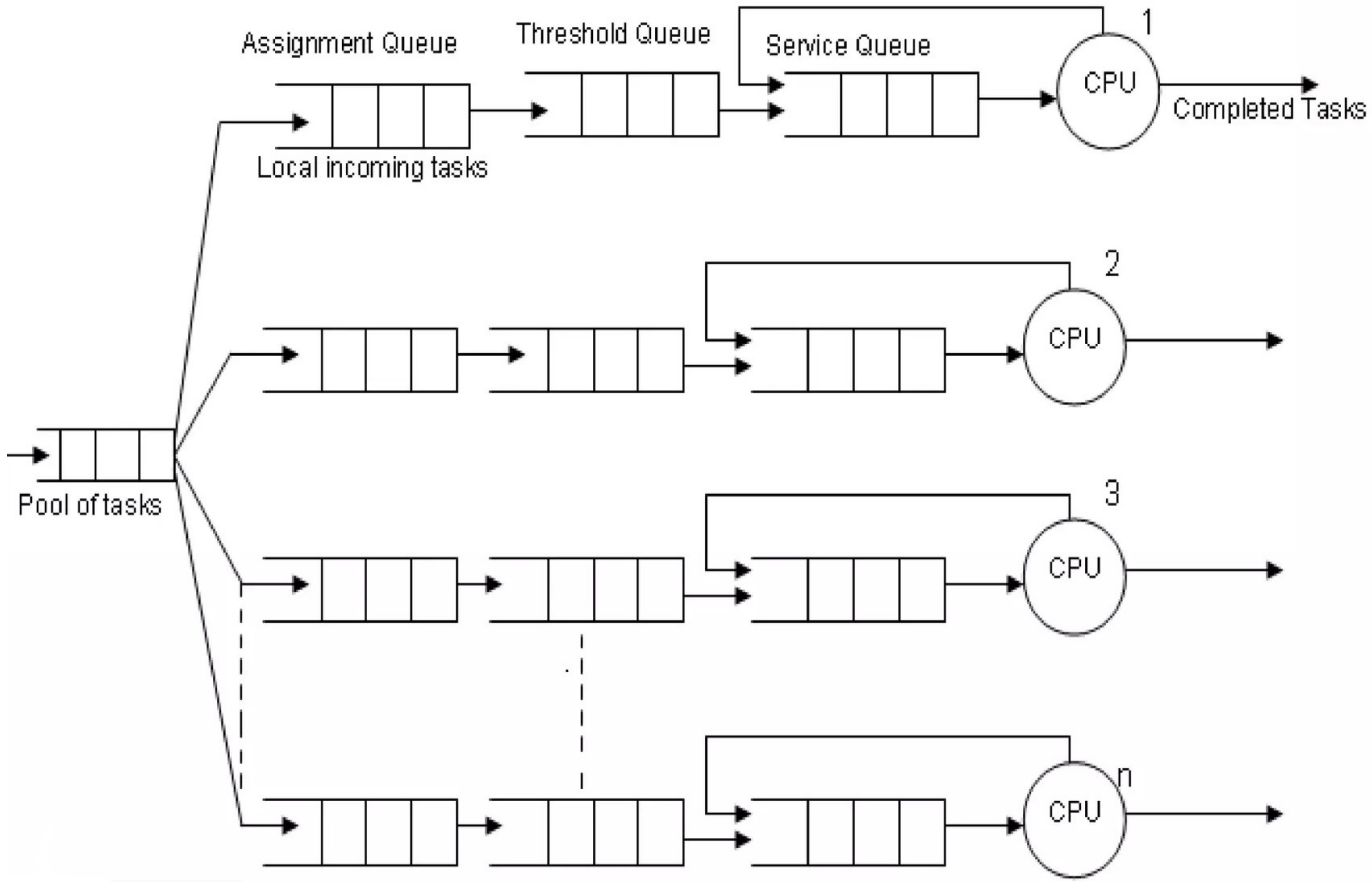


Figure : Model of Processing Node

Dynamic Load Balancing

- In **dynamic load balancing algorithm** assignment of jobs is done at the runtime.
- In DLB jobs are **reassigned at the runtime** depending upon the situation that is the load will be transferred from heavily loaded nodes to the lightly loaded nodes.
- In this case **communication over heads** occur and becomes more when number of processors increase.
- In dynamic load balancing no decision is taken until the process gets execution.

Cont...

- This strategy collects the information about the system state and about the job information.
- As more information is collected by an algorithm in a short time, potentially the algorithm can make better decision.
- Dynamic load balancing is mostly considered in heterogeneous system because it consists of nodes with different speeds, different communication link speeds, different memory sizes, and variable external loads due to the multiple.

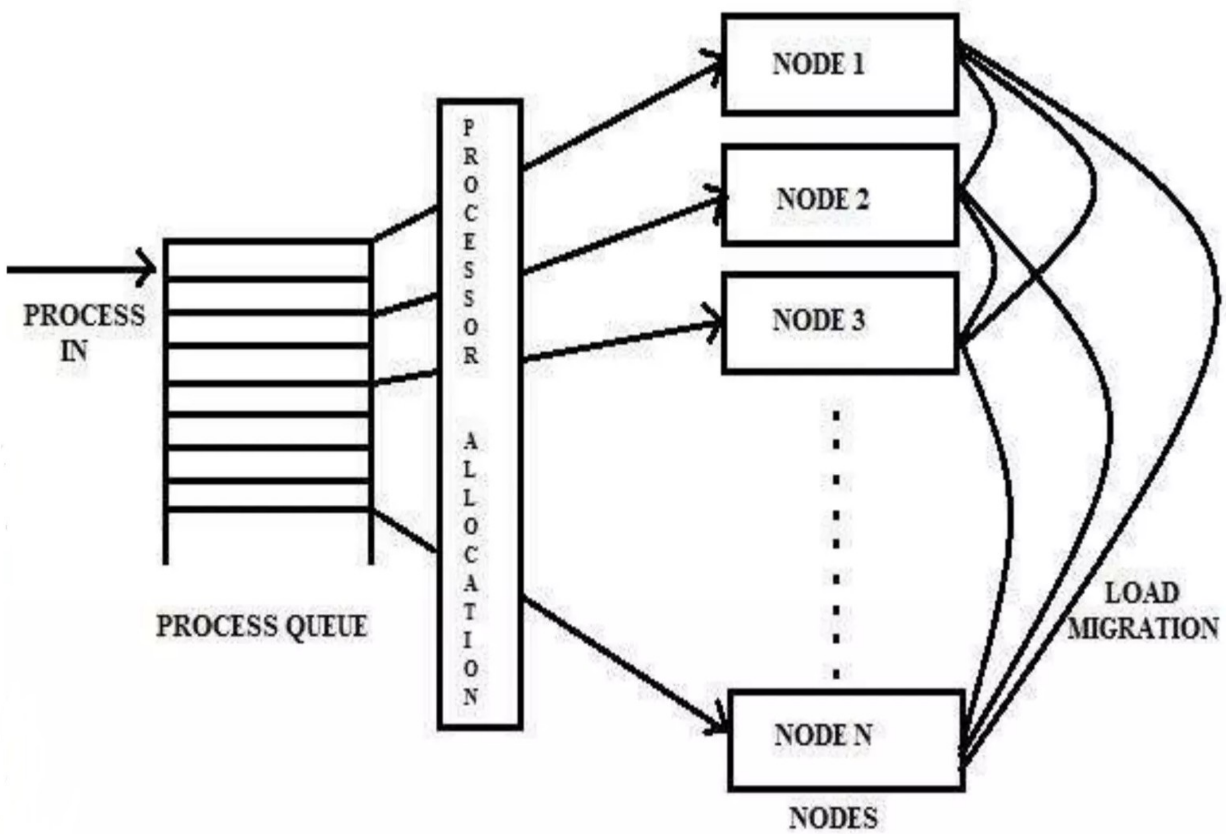


Figure: Job Migration in Dynamic Load Balancing Strategy

Advice

Kindly read the book before and after the lecture, if some terms/ words are not covered in the lecture, if appear in exam it might not appear strange or out of course.